

3CX Supply Chain Lab

Reconstruct the 3CX supply chain attack by analyzing compromised MSI and DLL artifacts to identify TTPs and attribute the incident to a threat actor.

Category:

[Threat Intel](#)

Tactics:

[Persistence](#)[Privilege Escalation](#)[Defense Evasion](#)[Discovery](#)

Tool:

[VirusTotal](#)

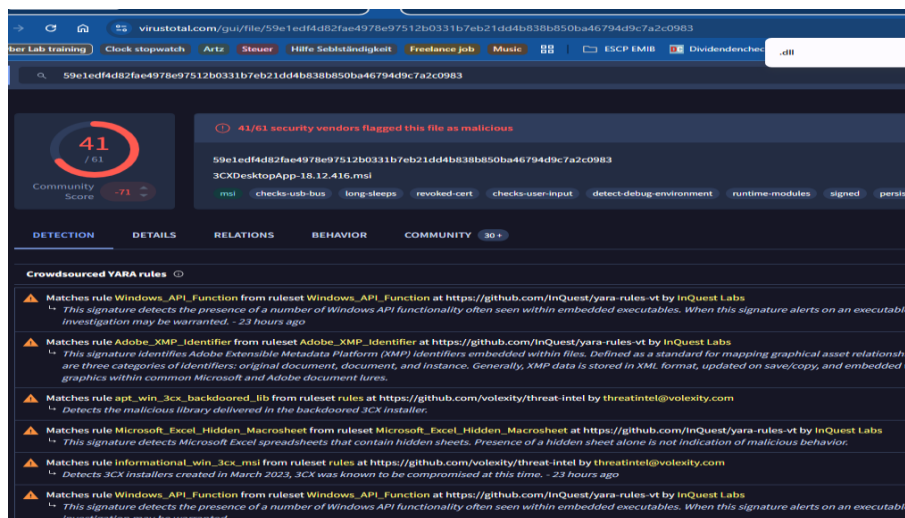
Scenario

A large multinational corporation heavily relies on the 3CX software for phone communication, making it a critical component of their business operations. After a recent update to the 3CX Desktop App, antivirus alerts flag sporadic instances of the software being wiped from some workstations while others remain unaffected. Dismissing this as a false positive, the IT team overlooks the alerts, only to notice degraded performance and strange network traffic to unknown servers. Employees report issues with the 3CX app, and the IT security team identifies unusual communication patterns linked to recent software updates.

As the threat intelligence analyst, it's your responsibility to examine this possible supply chain attack. Your objectives are to uncover how the attackers compromised the 3CX app, identify the potential threat actor involved, and assess the overall extent of the incident.

File Hash (SHA-256): 59e1edf4d82fae4978e97512b0331b7eb21dd4b838b850ba46794d9c7a2c0983

<https://www.virustotal.com/gui/file/59e1edf4d82fae4978e97512b0331b7eb21dd4b838b850ba46794d9c7a2c0983>



Q1
Solved : 4605

Understanding the scope of the attack and identifying which versions exhibit malicious behavior is crucial for making informed decisions if these compromised versions are present in the organization. How many versions of 3CX **running on Windows** have been flagged as malware?

2

Q2

Solved : 4485

Determining the age of the malware can help assess the extent of the compromise and track the evolution of malware families and variants. What's the UTC creation time of the .msi malware?

2023-03-13 06:33

Q3

Solved : 4325

Executable files (.exe) are frequently used as primary or secondary malware payloads, while dynamic link libraries (.dll) often load malicious code or enhance malware functionality. Analyzing files deposited by the Microsoft Software Installer (.msi) is crucial for identifying malicious files and investigating their full potential. Which malicious DLLs were dropped by the .msi file?

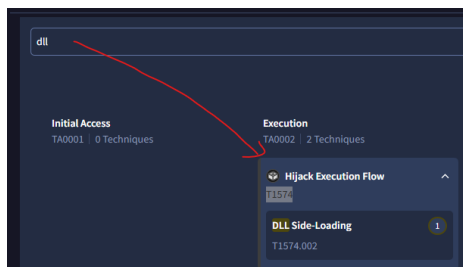
ffmpeg.dll,d3dcompiler_47.dll

Q4

Solved : 4289

Recognizing the persistence techniques used in this incident is essential for current mitigation strategies and future defense improvements. What is the MITRE Technique ID employed by the .msi files to load the malicious DLL?

T1574 , DLL side loading

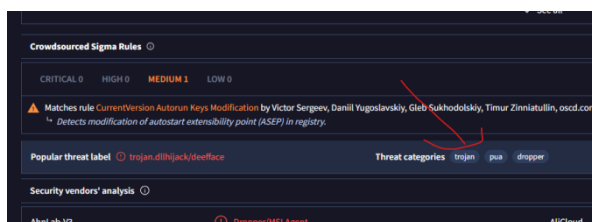


Q5

Solved : 4253

Recognizing the malware type (threat category) is essential to your investigation, as it can offer valuable insight into the possible malicious actions you'll be examining. What is the threat category of the two malicious DLLs?

Trojan

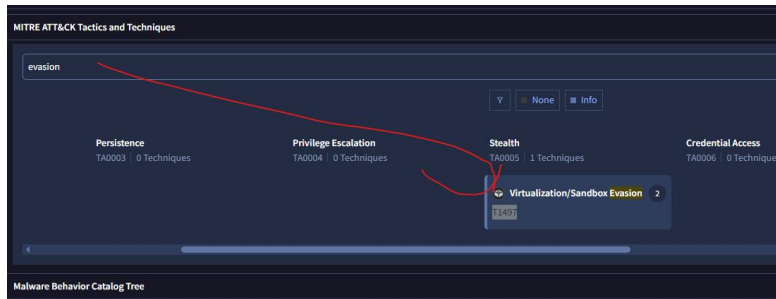


Q6

Solved : 4225

As a threat intelligence analyst conducting dynamic analysis, it's vital to understand how malware can evade detection in virtualized environments or analysis systems. This knowledge will help you effectively mitigate or address these evasive tactics. What is the MITRE ID for the virtualization/sandbox evasion techniques used by the two malicious DLLs?

T1497



Q7

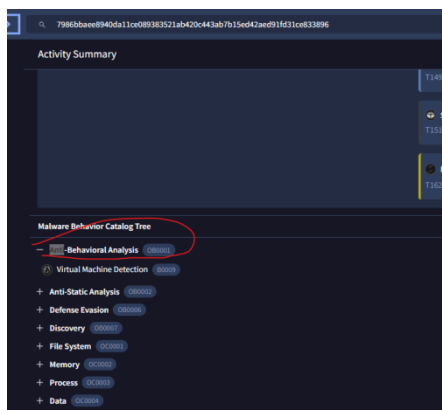
Solved : 4141

When conducting malware analysis and reverse engineering, understanding anti-analysis techniques is vital to avoid wasting time. Which hypervisor is targeted by the anti-analysis techniques in the ffmpeg.dll file?

In Virtualtotal

Behavior → Antivm / Anti-analysis section

VMware



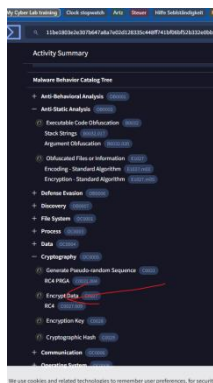
Q8

Solved : 4149

Identifying the cryptographic method used in malware is crucial for understanding the techniques employed to bypass defense mechanisms and execute its functions fully. What encryption algorithm is used by the ffmpeg.dll file?

Behavior → Process Tree / Memory Pattern / Execution Flow

RC4



Q9
Solved : 4128

As an analyst, you've recognized some TTPs involved in the incident, but identifying the APT group responsible will help you search for their usual TTPs and uncover other potential malicious activities. Which group is responsible for this attack?

Associated threat actor: APT29

Lazarus

Here's a concise table summarizing what the **3CX supply-chain malware (ffmpeg.dll MSI infection)** does:

One-line summary:

A supply-chain trojan that uses DLL sideloading + RC4 decryption + in-memory execution to turn a legitimate app into a stealth backdoor for espionage.

Stage	Component / Technique	What Happens	Purpose
1. Initial compromise	MSI installer (trojanized 3CX update)	Legitimate software update is modified to include malicious files	Initial infection via trusted software channel
2. Payload drop	ffmpeg.dll + d3dcompiler_47.dll	DLLs are placed alongside legitimate application binaries	Prepare environment for DLL sideloading
3. Execution trigger	DLL side-loading	Legitimate 3CX.exe loads malicious ffmpeg.dll instead of real library	Hijacks trusted process execution
4. Loader stage	ffmpeg.dll	Executes malicious logic instead of media functionality	Starts malware chain
5. Decryption	RC4-encrypted payload	Embedded payload is decrypted in memory	Hides malicious code from static analysis
6. In-memory execution	Shellcode / injected payload	Decrypted code runs directly in memory (fileless)	Avoids disk detection

Stage	Component / Technique	What Happens	Purpose
7. C2 communication	HTTPS / external domains	System fingerprint sent to attacker infrastructure	Enables remote control
8. Secondary payload	Infostealer/backdoor module	Steals browser data, credentials, system info	Espionage and data exfiltration
9. Persistence / follow-up	C2 commands	Receives instructions for further actions	Long-term access and control